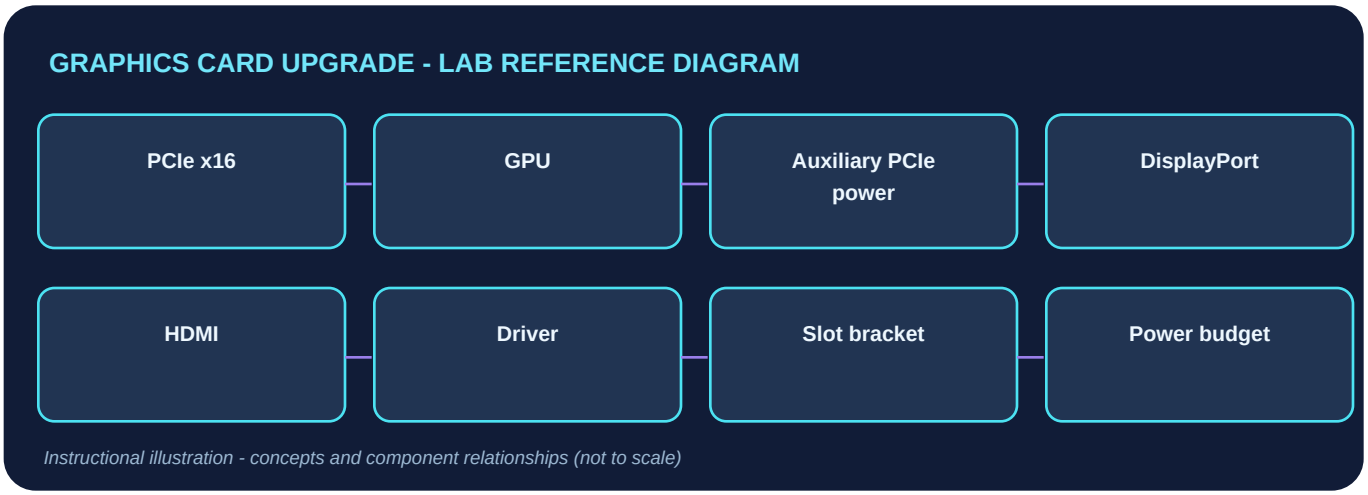
2.2.13 Upgrade a Video Card

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Install and verify a graphics adapter.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Confirm slot clearance and PSU wattage/connectors.
2. Power off and seat GPU in the correct PCIe slot.
3. Attach required PCIe power and display cable to GPU.
4. Install drivers and verify video operation.

Troubleshooting and safe practice

• Black screen: attach monitor to upgraded GPU.

• Instability: check power connector and PSU capacity.

2.2.13 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
PCIe x16	Expansion slot commonly used by graphics cards.
GPU	A key term in graphics card upgrade; identify its purpose, where it is used, and how to verify it.
Auxiliary PCIe power	Electrical power characteristic or component; assess safely with rated tools and correct connectors.
DisplayPort	Digital display interface common on computers and monitors.
HDMI	Digital connection carrying video and often audio.
Driver	Software enabling the operating system to communicate with hardware.
Slot bracket	Physical interface or connection point; verify its intended function and compatibility before use.
Power budget	Electrical power characteristic or component; assess safely with rated tools and correct connectors.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____